Highlights

Auditable decisions from unstructured text: typed language-model answers, DMN tables and risk-controlled human review

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- A typed probabilistic model is distilled into a verified DMN decision table.
- The model's probabilities drive a learned confidence gate expressed in DMN.
- Learn-then-Test bounds both the error rate and the critical-case miss rate.
- Exported decision models are XSD-valid and agree with an independent FEEL engine.
- Evaluated on three decisions over CFPB complaints and fraudulent job postings.

Auditable decisions from unstructured text: typed language-model answers, DMN tables and risk-controlled human review

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ABSTRACT

Operational decisions in business processes increasingly start from unstructured input, such as complaints, e-mails or claims, that rule-based decision models cannot read. Language models can read such input, but their decision logic cannot be audited, and it is unclear when their output may be acted on without a human. We study System One models, which answer a fixed set of typed questions (choices, yes/no statements, ordinal scores) with a probability distribution per answer, and show how to turn such a model into an auditable decision service based on the Decision Model and Notation (DMN) standard. The model serves twice: as a teacher, whose holistic decisions on unlabelled cases are distilled into a verified DMN decision table over its own typed answers, and as a source of uncertainty, whose probabilities feed a confidence gate. The gate's threshold is chosen with Learn-then-Test, so that, with high probability, both the error rate of automated cases and the share of critical cases, such as frauds, cleared automatically stay below a chosen level. We evaluate the approach with Jev, a commercial System One model, on three decisions: monetary relief and queue routing for 17,674 consumer complaints, and fraud screening for 14,562 job postings. We report the fidelity and size of the distilled tables, the calibration of the model, the share of cases that can be automated at error levels of 1–10%, the number of labelled critical cases such guarantees require, and the comparison with the model alone and with supervised baselines.

1. Introduction

Operational decisions, such as approving a refund, routing a request or escalating a claim, are where business processes create or destroy value at scale (Dumas, La Rosa, Mendling and Reijers, 2018). The Decision Model and Notation (DMN) standard (Object Management Group, 2021) separates such decisions from process control flow and expresses them as decision tables whose rules business analysts can read, test and change, and which process engines execute directly (Hasić, De Smedt and Vanthienen, 2018; Figl, Mendling, Tokdemir and Vanthienen, 2018). Decision tables have a well-defined semantics, so overlapping and missing rules can be detected automatically (Calvanese, Dumas, Laurson, Maggi, Montali and Teinemaa, 2016, 2018). These properties make DMN attractive wherever decisions must be explained to customers, auditors or regulators. In the European Union, for example, high-risk AI systems must be transparent to their deployers and allow effective human oversight (European Parliament and Council of the European Union, 2024).

Decision tables, however, can only read structured data. A growing share of process instances starts from unstructured input: a complaint narrative, an e-mail, a call transcript, a claim description. Machine learning, and recently large language models (LLMs), can read such input (Vidgof, Bachhofner and Mendling, 2023), but they put the decision logic in a black box, and their confidence statements are often poorly calibrated (Guo, Pleiss, Sun and Weinberger, 2017; Xiong, Hu, Lu, Li, Fu, He and Hooi, 2024). Organisations therefore face a choice between models that read unstructured input but cannot be audited, and rules that can be audited but cannot read the input. Neither tells them when a decision may be taken without a human.

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A new class of models changes the shape of this problem. *System One models* (TypeSafe AI, 2026c) do not generate text. They receive a state, for example a complaint, and a set of typed questions, and return for every question a probability distribution over a predefined set of answers: an option of a *choice*, a level of an ordinal *score*, or the probability that a statement is true (a *noul*). The first model of this class, Jev, evaluates all questions in parallel and, according to its vendor, is trained to return calibrated probabilities at a latency of tens to hundreds of milliseconds (TypeSafe AI, 2026c). Two properties make such models a natural match for DMN. Their answers are typed values from finite domains, which correspond directly to DMN data types. And they attach a probability to every answer, which can serve as an input to decision logic.

Existing work leaves this match unused. Decision discovery derives DMN models from historical, structured process data (Bazhenova, Buelow and Weske, 2016; De Smedt, Hasić, vanden Broucke and Vanthienen, 2019; Goossens, De Smedt and Vanthienen, 2024; Leewis, Smit and Versendaal, 2024); our earlier work translated decision trees learned from such data into DMN tables (Etinger, Simić and Buljubašić, 2019) and combined tree ensembles in a layered architecture that generates DMN models (Simić, Tanković and Etinger, 2020). Text-based approaches extract decision models from textual descriptions of decision logic (Goossens, De Smedt and Vanthienen, 2023), or let LLMs write DMN tables from given business rules (Cherednichenko and Maliarenko, 2025). None of them uses a model with typed probabilistic output both as the source of the decision logic and as the source of the signal that decides when that logic may run unattended.

We propose a method that does both. The System One model plays two roles. As a *teacher*, it answers a set of decomposed fact questions and one holistic decision question on a large pool of unlabelled cases; a decision tree learned on its typed fact answers, with its holistic decision as the target, is translated into a DMN decision table, simplified and verified. At run time the model is reduced to a feature extractor, and the decision itself is taken by the table, which can be read and audited. As a *source of uncertainty*, the model's probabilities feed a confidence gate, itself a small DMN table, that decides whether a case is decided automatically or sent to a person. The gate's threshold is chosen on a small labelled calibration set with Learn-then-Test (Angelopoulos, Bates, Candès, Jordan and Lei, 2025), which guarantees, with a chosen probability, that the error rate among automatically decided cases does not exceed a chosen level; for decisions with a rare, costly class, such as fraud, it also bounds the share of that class that is decided automatically and wrongly. Disagreement between the table and the model's holistic answer is used as an additional signal for review. The whole decision service is one decision requirements diagram that a BPMN process calls from a business rule task.

We address four research questions:

RQ1 How faithfully does a DMN table distilled from the model's typed answers reproduce the model's decisions, and at what table complexity?

RQ2 How well calibrated are the model's probabilities with respect to the true outcomes, and does post-hoc recalibration help?

RQ3 What share of cases can be decided automatically while the error rate of automated cases is kept below a given level?

RQ4 How does the approach compare with the model alone and with supervised baselines in accuracy, latency and cost, and how many labels would the supervised alternatives need?

We answer them on three decisions from two public data sets: whether a banking complaint to the Consumer Financial Protection Bureau (CFPB) should be resolved with monetary relief, to which of eight queues a complaint should be routed (17,674 complaints), and whether a job posting is fraudulent (14,562 postings of the EMSCAD corpus (Vidros, Kolias, Kambourakis and Akoglu, 2017)). The contributions of the paper are:

(1) a distillation method that turns the typed answers and holistic decisions of a System One model into DMN decision tables that are complete and free of conflicts by construction and that are verified statically, by round trip and by an independent FEEL engine;

(2) a confidence-gate pattern for DMN in which the gate's rules are learned from the model's probabilities and its threshold carries a joint finite-sample guarantee on the error rate of automated cases and on the miss rate of a critical class, together with the number of labelled critical cases this guarantee requires;

(3) an independent evaluation of the calibration and decision quality of a commercial System One model on three business decisions, which separates its value as a feature extractor from its value as a teacher, against supervised baselines and their labelling cost;

(4) an open implementation with executable DMN and BPMN models.

Section 2 reviews related work. Section 3 presents the method, Section 4 the experimental design and Section 5 the results. Section 6 discusses implications and limitations, and Section 7 concludes.

2. Background and related work

2.1. DMN decision tables

DMN defines decision requirements diagrams (DRDs), in which decisions depend on input data, other decisions and business knowledge models, and are governed by knowledge sources (Object Management Group, 2021). The most common decision logic is the decision table: each rule is a row of input entries, unary tests written in the Friendly Enough Expression Language (FEEL), and an output entry. A hit policy determines the result when several rules match; under the *unique* policy at most one rule may match any input. Calvanese et al. (2016, 2018) gave decision tables a formal semantics, interpreted rules as hyper-rectangles in the input space and derived algorithms that detect overlapping rules and gaps (inputs no rule covers) and that merge rules to simplify tables. Hasić et al. (2018) formulated principles for integrating decision and process models, and Figl et al. (2018) reviewed what is known about the understandability and use of DMN. We reuse the geometric view of Calvanese et al. (2018) to verify the tables we generate.

2.2. Discovering decision logic from data

Decision mining derives the conditions at process branching points from event logs (Rozinat and van der Aalst, 2006; de Leoni and van der Aalst, 2013). Later work produces DMN models: from process models in which decision logic is hard-coded (Batoulis, Meyer, Bazhenova, Decker and Weske, 2015), from event logs (Bazhenova et al., 2016), holistically from process execution data (De Smedt et al., 2019), and from object-centric process data (Goossens et al., 2024). Leewis et al. (2024) proposed a framework that organises the discovery of operational decisions from data. In our earlier work we translated scikit-learn decision trees learned on structured data into DMN tables (Etinger et al., 2019), and proposed a layered ensemble of decision-tree classifiers, generated as DMN models, to control the bias–variance trade-off on socially sensitive decision tasks (Simić et al., 2020). All of these approaches learn from structured attributes recorded in historical data, and their tables are only as good as the recorded labels.

Two lines of work start from text. Goossens et al. (2023) extract DMN models from natural-language descriptions of decision logic with deep learning, and Cherednichenko and Maliarenko (2025) let an LLM with retrieval write DMN tables from business rules. Arco, Nápoles, Vanhoenshoven, Lara, Casas and Vanhoof (2021) use natural-language processing to help decision modelers identify the inputs, outputs and rules in such descriptions. All of these translate a description of the logic into a model; none derives the logic from case data. Bork, Ali and Dinev (2023) argue for hybrid decision management that combines AI components with explicit decision models, and LLM-based generation of process models is surveyed by Horner, Moller and Reichert (2026). Our setting differs from all of these: the input of each case is unstructured text, the logic is learned from a model's decisions on cases, and the model's uncertainty becomes part of the decision model.

2.3. Distillation and interpretable surrogates

Knowledge distillation transfers the behaviour of a teacher model to a student (Hinton, Vinyals and Dean, 2015). When the student is a decision tree, the result is an interpretable surrogate of a black-box model (Craven and Shavlik, 1996; Bastani, Kim and Bastani, 2017; Frosst and Hinton, 2017); rule ensembles are an alternative (Friedman and Popescu, 2008). Surrogates are judged by their *fidelity*, the agreement with

the teacher, and by their complexity (Guidotti, Monreale, Ruggieri, Turini, Giannotti and Pedreschi, 2018). Rudin (2019) argues that in high-stakes settings the deployed model should itself be interpretable rather than explained after the fact, for example by local post-hoc explanations (Ribeiro, Singh and Guestrin, 2016); see Barredo Arrieta, Díaz-Rodríguez, Del Ser, Bennetot, Tabik, Barbado, Garcia, Gil-Lopez, Molina, Benjamins, Chatila and Herrera (2020) for a broader review of explainable AI. Our design follows that advice: the distilled table is the deployed decision logic, not an explanation of it.

The student in our method does not see the raw text but the teacher's answers to human-readable questions. This is the structure of concept bottleneck models, in which a model first predicts human-interpretable concepts and a simple model maps concepts to the output (Koh, Nguyen, Tang, Mussmann, Pierson, Kim and Liang, 2020). Recent work builds such bottlenecks with language models. Benara, Singh, Morris, Antonello, Stoica, Huth and Gao (2024) represent a text by an LLM's answers to a set of yes/no questions and fit sparse linear models on these answers, Sun, Oikarinen, Ustun and Weng (2025) train concept bottleneck LLMs for text classification, and Singh, Askari, Caruana and Gao (2023) use an LLM only during training to build fully transparent models such as linear models and trees over n-grams. Using a language model as an annotator to replace human labelling has been studied for text classification (Wang, Liu, Xu, Zhu and Zeng, 2021; Gilardi, Alizadeh and Kubli, 2023). We combine the two ideas: the System One model supplies both the concepts (typed facts) and the labels (holistic decisions), so no ground-truth labels are needed to build the table. Unlike the approaches above, the concepts arrive with probability distributions that a gate can use, and the student is a decision table in an industry standard rather than a linear layer.

2.4. Calibration, selective prediction and risk control

A model is calibrated when its predicted probabilities match observed frequencies. Calibration is assessed with the Brier score (BRIER, 1950), the expected calibration error (ECE) and reliability diagrams (Pakdaman Naeini, Cooper and Hauskrecht, 2015; Guo et al., 2017), and can be improved post hoc, for example with isotonic regression (Zadrozny and Elkan, 2002). Whether LLMs know what they know has been studied for token probabilities and verbalised confidence (Kadavath et al., 2022; Tian, Mitchell, Zhou, Sharma, Rafailov, Yao, Finn and Manning, 2023; Xiong et al., 2024); Geng, Cai, Wang, Koeppl, Nakov and Gurevych (2024) survey confidence estimation and calibration for LLMs, and Kapoor, Gruver, Roberts, Collins, Pal, Bhatt, Weller, Dooley, Goldblum and Wilson (2024) show that prompting alone does not produce reliable uncertainty estimates, whereas fine-tuning on a small set of graded answers does; System One models are claimed to be trained for calibration directly (TypeSafe AI, 2026c), which we test independently.

Selective prediction lets a classifier abstain. Chow (1970) characterised the optimal trade-off between error and rejection; Geifman and El-Yaniv (2017) gave a method to select a confidence threshold with a guaranteed bound on the selective risk. Learning to defer extends the idea to deferring to a human expert (Madras, Pitassi and Zemel, 2018; Mozannar and Sontag, 2020; Raghu, Blumer, Corrado, Kleinberg, Obermeyer and Mullainathan, 2019), and Wen, Yao, Feng, Xu, Tsvetkov, Howe and Wang (2025) survey when and how LLMs should abstain. Conformal prediction gives distribution-free guarantees from exchangeable calibration data (Vovk, Gammerman and Shafer, 2022); conformal risk control and Learn-then-Test (LTT) generalise these guarantees to arbitrary risks and to the choice of any finite set of parameters by multiple hypothesis testing (Angelopoulos, Bates, Fisch, Lei and Schuster, 2024; Angelopoulos et al., 2025). For language models, Quach, Fisch, Schuster, Yala, Sohn, Jaakkola and Barzilay (2024) calibrate conformal sets of generated answers, and Ren, Dixit, Bodrova, Singh, Tu, Brown, Xu, Takayama, Xia, Varley, Xu, Sadigh, Zeng and Majumdar (2023) use conformal prediction to decide when an LLM-based planner should ask a person for help, which is the role our gate plays in a business process. We use LTT with fixed-sequence testing because the parameter we tune, the gate threshold, has a natural order and the controlled quantity, the error rate among automated cases, is what a process owner can specify.

2.5. System One models

The term follows the distinction between fast, intuitive judgements (System 1) and slow, deliberate reasoning (System 2) (Kahneman, 2011). A System One request contains a *state* (text or JSON) and a map of questions. A *choice* selects one of up to 255 options and returns a distribution over them; a *score* rates the state on an ordered rubric of up to ten levels and returns a distribution over levels and its expected value; a *noul* returns the probability that a statement is true. Every question is answered in isolation from

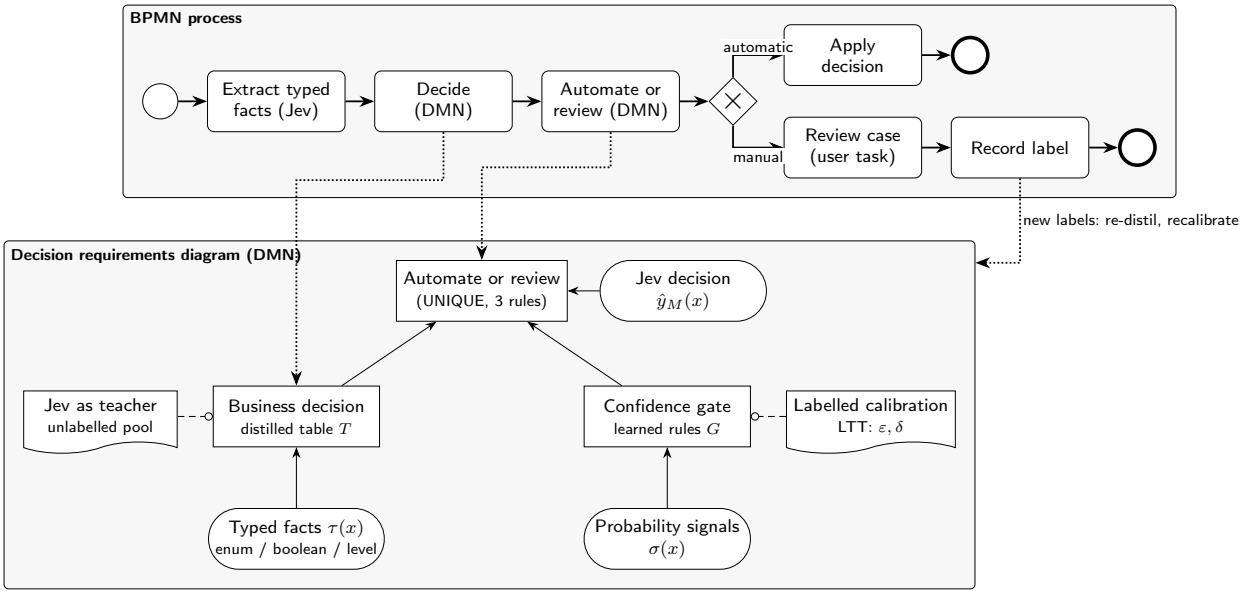


Figure 1: Architecture. The BPMN process calls the System One model once per case and then evaluates a decision requirements diagram with three decisions. The business decision table is distilled from the model's own decisions (knowledge source on the left); the confidence gate is learned and thresholded on a labelled calibration set (knowledge source on the right). Cases decided by a reviewer produce labels for re-distillation and recalibration.

Table 1
Mapping of System One answer types to DMN. p denotes the probability of the most likely answer; n the number of options; $m^\star$ the most likely score level and $\bar{d}$ the mean absolute deviation of a uniform distribution over the levels.

Question	Model output	Typed value $\tau_j(x)$	DMN type	Signals in $\sigma(x)$
Choice	distribution over $n \leq 255$ options	most likely option	string, enumerated	p, confidence $\frac{p-1/n}{1-1/n}$, margin to 2nd
Noul	probability of "true"	$p \geq 0.5$	boolean	p, confidence $\lvert 2p-1 \rvert$
Score	distribution over $L \leq 10$ levels	level $m^\star$	number in $\{0, \ldots, L-1\}$	p, confidence $1 - \frac{\sum_\ell \pi_\ell \lvert \ell - m^\star \rvert}{\bar{d}}$, margin

the others, in parallel, and answers can only take the declared values. The model we use, Jev, reports a confidence for choices and scores that rescales the top probability (choices) or the spread around the mode (scores) to $[0, 1]$ (TypeSafe AI, 2026c). The vendor documents known failure modes, among them literal reading of questions, weak arithmetic and date handling, and a preference for the first option of a choice (TypeSafe AI, 2026a); our question design and a robustness check address these (Sections 4.2 and 5.4).

3. Method

3.1. Overview and notation

Figure 1 shows the architecture. A case x, here a complaint narrative, enters a BPMN process. A service task sends x to the System One model M with a fixed schema of m *fact questions* $q_1, \ldots, q_m$ and one *holistic question* h per decision. For each fact question q_j with answer domain D_j, M returns a distribution $\pi_j(x)$ over D_j; for h it returns a distribution $\pi_h(x)$ over the decision classes C. A business rule task then evaluates a DRD with three decisions: a *business decision* table T over the typed facts, a *confidence gate* G over probability signals, and an *outcome* decision that automates the case only if the gate accepts it and T agrees with M's holistic answer. Otherwise a user task asks a person to decide, and the person's decision is stored as a new label.

Algorithm 1: Distillation of a DMN decision table from a System One model.

Input: pool U; schema $(q_1, \ldots, q_m, h)$; grids of depths and minimum leaf sizes; tolerance t
Output: decision table T with hit policy UNIQUE

Query M once per $x \in U$; obtain $\tau(x)$ and $\hat{y}_M(x)$;
Encode $\tau(x)$: one-hot for choices, 0/1 for nouls, integer level for scores;
Split U into U_{fit} (80%) and U_{val} (20%);
foreach *configuration* (d, ℓ) *in the grid* **do**
    Fit a CART tree with balanced class weights, depth $\leq d$ and leaf size $\geq \ell$ on U_{fit} with target $\hat{y}_M$;
    $T_{d,\ell} \leftarrow$ MERGE(PATHS(tree)); record $|T_{d,\ell}|$, balanced fidelity $F_{d,\ell}$ on U_{val} and its standard error $se_{d,\ell}$;
$(d^*, \ell^*) \leftarrow \arg\max F_{d,\ell}$; select the configuration with the fewest rules among those with $F_{d,\ell} \geq F_{d^*,\ell^*} - \max(t, se_{d^*,\ell^*})$;
Refit it on U; **return** $T \leftarrow$ MERGE(PATHS(tree)) without unused inputs;

Table 1 gives the mapping from answers to DMN inputs. The typed value $\tau_j(x)$ is the most likely answer; together the typed values form the vector $\tau(x) \in D = D_1 \times \cdots \times D_m$. The signals $\sigma(x)$ summarise how certain each answer is; the confidence formulas are those of the vendor (TypeSafe AI, 2026c). The model's holistic decision is $\hat{y}_M(x) = \arg\max_{c \in C} \pi_h(x)_c$ and the ground-truth decision, where available, is $y(x)$.

3.2. Distilling the business decision table

The aim is a table $T : D \to C$ that reproduces $\hat{y}_M$ as closely as possible with as few rules as possible. Its *fidelity* is $F(T) = \Pr[T(\tau(x)) = \hat{y}_M(x)]$. Distillation needs no ground-truth labels: a pool U of unlabelled cases is sent to M once, and the pairs $(\tau(x), \hat{y}_M(x))$, $x \in U$, are the training data. Algorithm 1 summarises the procedure.

From paths to rules. PATHS turns each root-to-leaf path into one rule. Every split refers to an encoded column, which is mapped back to a condition on the original typed input: a split on the one-hot column for option o of choice q_j yields $\tau_j \in \{o\}$ or $\tau_j \in D_j \setminus \{o\}$; a split on a noul yields `true` or `false`; a split on a score yields a range of levels. Conditions on the same input are intersected along the path, so each rule assigns to every input a set $S_j \subseteq D_j$, written in FEEL as a list of values, a negated list (`not("a","b")`), a range or `-` when $S_j = D_j$. The rule's output is the majority class of its leaf; its support and purity are kept as rule annotations. Because the leaves of a tree partition the encoded space and every element of D has exactly one encoding, the resulting rules partition D:

Lemma 1. *The rules produced by PATHS are pairwise disjoint and cover D. Hence the table satisfies the UNIQUE hit policy and has no gaps on the whole input domain, not only on the observed cases.*

The same translation applies to the gate, whose inputs are real-valued probabilities. A split $v \leq \theta$ found on training values separates the largest value $a \leq \theta$ from the smallest value $b > \theta$; any threshold $c \in [a, b)$ induces the same partition of the training data. We therefore write the shortest decimal in $[a, b)$, so that the table shows `>= 0.88` rather than `> 0.87645`, without changing any decision on the data the gate was fitted on.

Merging. Two rules with the same output whose conditions coincide on all inputs but one, j, are replaced by a single rule with condition $S_j \cup S'_j$ (for real inputs only when the two intervals are adjacent, so that every cell remains one interval). As the two rules are disjoint boxes that differ in one dimension, their union is again a box and the table's semantics is unchanged; this is the merge step of Calvanese et al. (2018). We repeat merging to a fixpoint, grouping rules by their output and the conditions on all other inputs so that each pass is linear in the number of rules. Inputs that are `-` in every rule are removed. For comparison we also report the size of an equivalent table with hit policy FIRST, in which the rules of the most frequent output are replaced by a final catch-all rule.

Model selection. Fidelity usually grows with the number of rules while readability falls. Rather than fixing a size, we sweep tree depth and minimum leaf size and apply the one-standard-error rule of CART: we select

the smallest table whose validation fidelity is within one standard error of the best, and at least within $t = 1$ percentage point; the full fidelity–complexity curve is reported. Fidelity is measured as *balanced* agreement, the mean over classes of the share of the model's decisions of that class that the table reproduces, and trees are fitted with class weights inversely proportional to class frequency. Without both measures a rare decision, such as a fraud verdict given to a few percent of cases, is absorbed by a one-rule table that always returns the majority class and still agrees with the model on most cases.

3.3. Verification

Lemma 1 holds by construction, but the exported artefact must also be checked. We verify each table in four ways. First, we compute all pairs of overlapping rules and the uncovered part of D with box algebra: a rule is a box $\prod_j S_j$, two rules overlap if their conditions intersect on every input, and gaps are obtained by subtracting each rule from D, where the difference of two boxes is a union of at most m disjoint boxes (Calvanese et al., 2018). Second, the DRD is serialised to DMN 1.3 XML and validated against the OMG XML schema. Third, the XML is parsed back and its FEEL entries are evaluated by our own interpreter on every case; the outputs must equal those of the fitted tree. Fourth, an independent engine evaluates the same file: the DMN is parsed with `dmn-moddle` and its expressions and unary tests are evaluated with the FEEL interpreter `feelin` from the bpmn.io tool chain (feelin contributors, 2026). Agreement of the four is required before a model is deployed.

3.4. Risk-controlled confidence gate

Let $c(x) = \mathbb{1}[T(\tau(x)) = y(x)]$ indicate that the table decides case x correctly and $a(x) = \mathbb{1}[T(\tau(x)) = \hat{y}_M(x)]$ that the table agrees with the model's holistic answer. A gate is a score $s(x)$ together with a threshold λ; a case is automated when $\tilde{s}(x) \geq \lambda$, where $\tilde{s}(x) = s(x)$ if $a(x) = 1$ and $-\infty$ otherwise, so that disagreement always leads to review. The quantity a process owner cares about is the *selective risk*

$$R(\lambda) = \Pr\left[c(x) = 0 \,\middle|\, \tilde{s}(x) \geq \lambda\right], \tag{1}$$

the error rate among automated cases, and the *coverage* $\Pr[\tilde{s}(x) \geq \lambda]$, the share of cases automated. When one class $c^\star$ is rare and costly to miss, R alone is not informative: if 4% of job postings are fraudulent, automating every posting as legitimate already satisfies $R \leq 0.05$. For binary decisions we therefore designate the positive class as critical (refund owed; fraudulent posting) and also control the *miss rate*

$$M(\lambda) = \Pr\left[\tilde{s}(x) \geq \lambda,\ c(x) = 0 \,\middle|\, y(x) = c^\star\right], \tag{2}$$

the share of critical cases that are decided automatically and wrongly. Given a tolerated level ε and a failure probability δ, we want the largest coverage such that $R(\lambda) \leq \varepsilon$, and $M(\lambda) \leq \varepsilon$ where a critical class exists, hold jointly with probability at least $1 - \delta$.

Scores. We compare two scores. The first is the model's own confidence in its holistic answer, $s(x) = \max_c \pi_h(x)_c$. The second is a *learned gate*: a shallow classification tree (depth ≤ 3, at least 40 cases per leaf) fitted on the signals $\sigma(x)$ of the holistic question and of the facts used by T, with target $c(x)$; its score is the leaf estimate of $\Pr[c(x) = 1]$. When a critical class exists, its cases receive weight $n_{\neg c^\star}/n_{c^\star}$ in the fit, so that the leaves separate the cases whose automated error would be a miss rather than only the frequent ones. Because the tree is shallow, its leaves translate into a handful of readable DMN rules such as `relief_holistic__conf >= 0.88`.

Threshold. The labelled calibration set is split at random into halves A and B. Half A is used to fit the learned gate and to fix an ordered list of candidate thresholds $\lambda_1 > \lambda_2 > \cdots > \lambda_K$ (Algorithm 2); half B is used only for testing. For each candidate, let n_k be the number of cases in B with $\tilde{s} \geq \lambda_k$ and e_k the number of errors among them. The null hypothesis $H_k : R(\lambda_k) > \varepsilon$ is tested with the exact binomial p-value $p_k = \Pr[\mathrm{Bin}(n_k, \varepsilon) \leq e_k]$, and the hypotheses are tested in order until the first non-rejection. With a critical class, a second sequence tests $H'_k : M(\lambda_k) > \varepsilon$ on the same candidates with $p'_k = \Pr[\mathrm{Bin}(n^\star, \varepsilon) \leq e^\star_k]$, where $n^\star$ is the number of critical cases in B and $e^\star_k$ the number of them automated wrongly at λ_k. Each sequence is run at level $\delta/2$ and the more conservative of the two thresholds is used.

Algorithm 2: Gate threshold by Learn-then-Test with fixed-sequence testing.

Input: scores and correctness on halves A and B; risk level ε; failure probability δ; $K_{\max}$
Output: threshold $\hat{\lambda}$ (accept $\tilde{s} \geq \hat{\lambda}$)

$n_{\min} \leftarrow \lceil \log \delta / \log(1-\varepsilon) \rceil$; `// fewest cases that can reject`

On A: order distinct finite scores decreasingly; start at the first λ for which a binomial test on A rejects H with at least $n_{\min}|A|/|B|$ accepted cases (else at the first with that many cases); keep $K_{\max}$ evenly spaced thresholds $\lambda_1 > \cdots > \lambda_K$;

$\hat{\lambda} \leftarrow +\infty$;

for $k = 1, \ldots, K$ **do**

  On B: $n_k \leftarrow |\{\tilde{s} \geq \lambda_k\}|$, $e_k \leftarrow$ errors among them, $p_k \leftarrow F_{\mathrm{Bin}(n_k,\varepsilon)}(e_k)$;

  if $p_k > \delta$ **then break**;

  $\hat{\lambda} \leftarrow \lambda_k$;

return $\hat{\lambda}$;

Proposition 1. *Assume that the cases of half B and future cases are independent and identically distributed and independent of half A and of the pool used to build T. Then the threshold $\hat{\lambda}$ of Algorithm 2 satisfies* $\Pr\left[R(\hat{\lambda}) \leq \varepsilon\right] \geq 1-\delta$. *With a critical class and both sequences run at $\delta/2$,* $\Pr\left[R(\hat{\lambda}) \leq \varepsilon \text{ and } M(\hat{\lambda}) \leq \varepsilon\right] \geq 1-\delta$.

Proof sketch. Conditional on A and the pool, the table, the score and the candidate list are fixed, so the hypotheses $H_1, \ldots, H_K$ are fixed. Conditional on n_k, the cases of B with $\tilde{s} \geq \lambda_k$ are an i.i.d. sample from the distribution of cases given $\tilde{s} \geq \lambda_k$, hence $e_k \sim \mathrm{Bin}(n_k, R(\lambda_k))$ and, under H_k, $\Pr[p_k \leq u \mid n_k] \leq u$ for all u; the p-value is therefore valid also unconditionally. Fixed-sequence testing rejects a true null only if the first true null in the sequence is rejected, which happens with probability at most δ whatever the dependence among the p-values; it thus controls the family-wise error rate at δ (Angelopoulos et al., 2025). If $R(\hat{\lambda}) > \varepsilon$, the true null $H_{\hat{k}}$ has been rejected, so $\Pr[R(\hat{\lambda}) > \varepsilon] \leq \delta$. For the miss rate, the $n^\star$ critical cases of B are i.i.d. draws from the class-conditional distribution and $e_k^\star \sim \mathrm{Bin}(n^\star, M(\lambda_k))$, so the same argument applies; M is non-increasing in λ, so the conservative-first order is also the natural testing order. Fixed-sequence testing controls the family-wise error rate, so with probability at least $1-\delta/2$ every rejected H_k is false, and likewise for every rejected H'_k. The selected threshold is the more conservative of the two last rejections and therefore lies in both rejected prefixes; by the union bound over the two sequences, $R(\hat{\lambda}) \leq \varepsilon$ and $M(\hat{\lambda}) \leq \varepsilon$ hold jointly with probability at least $1-\delta$. The argument does not require R to be monotone in λ. □

The guarantee does not require the model's probabilities to be calibrated; calibration only affects how many cases can be automated. Choosing the start of the sequence on A matters in practice: fixed-sequence testing stops at the first non-rejection, and a sequence that starts at a threshold accepting only a few cases would stop immediately. When no candidate is rejected, $\hat{\lambda} = +\infty$ and every case is reviewed, which is the safe default. The miss-rate test also has a hard sample-size requirement: even with no misses, H'_1 can only be rejected if $(1-\varepsilon)^{n^\star} \leq \delta/2$, that is, if half B contains at least $n^\star_{\min} = \lceil \log(\delta/2)/\log(1-\varepsilon) \rceil$ critical cases (36 for $\varepsilon = 0.10$, 72 for 0.05, 183 for 0.02 and 368 for 0.01 at $\delta = 0.05$). This turns a risk policy into a labelling requirement that a process owner can plan for.

Gate and outcome tables. The learned gate is translated with PATHS; each leaf outputs `"auto"` if its score is at least $\hat{\lambda}$ and `"review"` otherwise, and the rules are merged. The outcome decision is a three-rule UNIQUE table with inputs *gate* and the FEEL expression `business_decision = holistic_decision`; it returns `"automatic"` only for (`"auto"`, `true`).

3.5. Deployment and maintenance

The DRD of Figure 1 is deployed together with a BPMN process for Camunda 8: a service task calls the model and writes typed facts, signals and the holistic answer into process variables, two business rule tasks call the business and outcome decisions, and an exclusive gateway routes the case. Two knowledge sources document where the rules come from (the model version and the pool size) and how the gate was

Table 2
Decision tasks. Ground truth comes from fields recorded by the CFPB and from the EMSCAD labels; the model sees only the text.

Task	Cases and classes	Ground truth
Monetary relief	8,000 random complaints about checking or savings accounts, credit and prepaid cards and money transfers; classes `relief`, `no_relief` (16.8% relief)	company response "Closed with monetary relief" versus "Closed with explanation" or "Closed with non-monetary relief"
Routing	10,000 complaints, 1250 per queue; eight queues: credit reporting, debt collection, deposits, cards, payments, mortgage, consumer loans, fraud	product group selected by the consumer; complaints whose issue or sub-issue denotes fraud, identity theft or unauthorised activity go to the fraud queue (14 issue/sub-issue pairs, listed in the repository)
Job-posting fraud	all 14,562 deduplicated EMSCAD postings; classes `fraudulent`, `legitimate` (4.2% fraudulent)	label assigned by the platform's staff (Vidros et al., 2017)

calibrated (ε, δ, size of the calibration set). Because a model alias may move to a new version, the request pins a versioned model identifier and the response's model field is logged with every decision. Decisions made by reviewers become labels; they are used to monitor the selective risk in production and to recalibrate the gate or re-distil the table when the model, the case mix or the policy changes.

4. Experimental design

4.1. Cases and data

We use two public data sets. The first is the CFPB Consumer Complaint Database (Consumer Financial Protection Bureau, 2026), in which companies record how they closed each complaint; its narratives have been analysed, for example, with topic models to support complaint handling (Bastani, Namavari and Shaffer, 2019). The CFPB stopped publishing complaint narratives on 14 September 2026 (Ballard Spahr LLP, 2026); we therefore take narratives from a public mirror of the database with complaints up to 2 January 2025 (Heineman, 2025) and join them by complaint identifier with the official bulk snapshot of 6 October 2026, which supplies the final company response. Only 1.32% of the joined complaints changed their response between the two snapshots, mostly from "In progress" to a final response. We keep complaints received between 1 April 2023 and 31 December 2024 with a narrative of at least 30 words, a closed response and a product in one of the ten product categories listed in Table 2, and drop exact duplicates of normalised narratives, which are common among templated letters and would otherwise leak between splits. This leaves 540,782 complaints.

The second is the Employment Scam Aegean Dataset (EMSCAD) (Vidros et al., 2017): 17 880 job postings published on a recruiting platform in 2012–2014, of which 866 were labelled fraudulent by the platform's staff. Its labels come from a dedicated review rather than from the parties of a dispute, which makes it a useful contrast to the CFPB labels. The original site of the corpus is no longer online; we use a public copy whose counts match the published ones and record its checksum. The model sees the posting as a short document with the title, location, salary, employment type, company profile, description, requirements and benefits; the platform's metadata (company logo, screening questions, telecommuting flag) are withheld, as the structured CFPB fields are. Postings with fewer than 30 words and postings whose description duplicates an earlier one, often the same posting for several locations, are removed, leaving 14,562 postings of which 4.2% are fraudulent.

From this corpus we draw two samples with a fixed seed (Table 2). The relief task is a real business decision taken by the company after handling the complaint; among the 106,092 banking complaints of the corpus, 17.0% were resolved with monetary relief. The routing task uses the consumer's own classification as a proxy for the correct handling queue, simulating an intake channel such as e-mail or a call centre where

Table 3
Question schema (abridged wording; full text in the repository). C = choice, N = noul, S = score.

Id	Type	Question
`product`	C (11)	Which financial product or service is the complaint mainly about? (ten products, "other")
`unauthorized`	N	Someone else made a transaction, charge or withdrawal, or opened an account, without the consumer's permission; disputes about the consumer's own purchases do not count.
`identity_theft`	N	The consumer is a victim of identity theft.
`scam`	N	A scammer tricked the consumer into sending money or giving account access.
`report_dispute`	N	The consumer disputes information on their credit report.
`collector_contact`	N	The consumer complains about contact from a debt collector.
`money_lost`	N	The consumer lost money or was charged money they believe they should not have paid.
`fee_or_interest`	N	The complaint is about a fee, a penalty or an interest charge.
`refund_requested`	N	The consumer asks for a refund, reimbursement, credit or return of money.
`prior_contact`	N	The consumer contacted the company about the problem before complaining.
`company_admitted`	N	The company admitted a mistake or promised to fix the problem.
`claim_denied`	N	The company denied or rejected the consumer's dispute, claim or request.
`funds_blocked`	N	The consumer cannot use their account or money because the company froze, held, restricted or closed it.
`harm`	S (4)	How much financial harm is described? (none; small; significant or ongoing; severe)
`relief_holistic`	N	Based on the complaint, the company should refund or reimburse money to the consumer.
`queue_holistic`	C (8)	Which team should handle this complaint? (eight queues with one-line descriptions)

no category is selected; the fraud queue makes the routing depend on more than the product. The union of both samples has 17,674 complaints with a median length of 170 words (interquartile range 94–289); 113 narratives longer than 1,500 words are truncated. Each task is split, stratified by its ground truth, into a pool (60%), a calibration set (20%) and a test set (20%): 4,799/1,600/1,601 cases for relief and 6,000/2,000/2,000 for routing. The EMSCAD postings are split in the same way (8,736/2,912/2,914). Ground truth of the pool is never used by the distilled table.

4.2. Question schema

All questions of the two CFPB tasks are asked in a single request per complaint, with the narrative as the state. The schema (Table 3) has fourteen fact questions and two holistic questions. The facts describe the complaint, not the decision: what product it concerns, whether the consumer reports unauthorised activity, a loss of money, a refund request or a denied claim, and how much financial harm is described. Following the vendor's guidance (TypeSafe AI, 2026a), every question states one literal condition, avoids arithmetic and dates and adds an explicit "other" option where the answer may not exist. Neither holistic question mentions the facts, so that the distilled table has to discover how the model's holistic judgement relates to them. The EMSCAD schema has fourteen fact questions on cues reported for employment scams (Vidros et al., 2017): the job category (choice), whether the company is described and named, remote work, stated pay, promises of easy money, no experience required, listed requirements and benefits, requests for payment or personal data, contact outside the platform (nouls), and the specificity of the duties and the quality of the writing (scores); its holistic noul states that the posting is a scam or a fake job.

Each schema was piloted once on 500 random pool cases (for EMSCAD also on 150 fraudulent pool cases, because a random sample contains too few) and then frozen; the file hashes are recorded with the data. The pilot looked for answers that are always or never true, unused options and a preference for the first option. One CFPB question was reworded: `unauthorized` was true for 37% of the pilot complaints because the model counted disputes about purchases the consumer had made; the final wording excludes them (26%). Three EMSCAD nouls (payment required, personal data requested, contact outside the platform) were almost never true, because the frauds in EMSCAD are mostly fake postings rather than advance-fee scams; they were kept unchanged, since rewording them towards the labels would fit the schema to the data. Pilot cases are pool cases, so no calibration or test case influenced the schemas.

4.3. Model and protocol

We call Jev through the System One API with the pinned model `jev-1.13.0` (reported model: jev-1.13.0). Each complaint is sent once; the request, the raw response, the latency and the time stamp are stored, and

all analyses read from this record. Decision trees use scikit-learn (Pedregosa, Varoquaux, Gramfort, Michel, Thirion, Grisel, Blondel, Prettenhofer, Weiss, Dubourg, Vanderplas, Passos, Cournapeau, Brucher, Perrot and Duchesnay, 2011) with depths $\{1, \ldots, 6, 8, 10, 12, \text{unbounded}\}$ and minimum leaf sizes $\{5, 20, 50, 100\}$. The gate is calibrated for $\varepsilon \in \{0.01, 0.02, 0.05, 0.10\}$ with $\delta = 0.05$; the gate exported to DMN uses $\varepsilon = 0.05$.

4.4. Baselines

We compare the distilled table (*Jev+DMN, distilled*; no ground-truth labels) with: the majority class; Jev's holistic answer (*Jev holistic*; no labels); a decision table learned with the same procedure on Jev's typed facts but with ground-truth labels as target (*Jev+DMN, supervised*, DMN-Sup), which separates Jev's value as a feature extractor from its value as a teacher and is also gated; and a TF-IDF representation of the narrative (word uni- and bigrams) with logistic regression (*TF-IDF+LR*), a strong supervised text classifier without auditable logic. The supervised baselines are trained on labelled subsets of the pool of $\{100, 250, 500, 1000, 2000\}$ cases (five random repetitions) and on the whole pool, which shows how many labels they need to match the label-free approach. For the gate we compare no gate, the agreement filter alone, Jev's confidence and the learned gate, each with and without the agreement filter.

4.5. Metrics and statistical analysis

Fidelity is the agreement between the table and Jev's holistic answer on the test set, with Cohen's κ (Cohen, 1960) and a 95% percentile-bootstrap interval (2,000 resamples). Accuracy, balanced accuracy (mean recall over classes) and, for binary tasks, recall and precision of the positive class are computed against the ground truth; balanced accuracy is the primary measure because two of the three tasks are imbalanced. Calibration is measured by the top-label ECE with 15 equal-width bins (and equal-mass bins), the maximum calibration error, the multi-class Brier score (BRIER, 1950) and reliability diagrams; isotonic recalibration is fitted on the calibration set and evaluated on the test set. For the gate we report coverage, selective risk and, for binary tasks, the miss rate on the test set, the area under the risk–coverage curve (AURC), and an empirical check of Proposition 1: the calibration and test cases are pooled and repeatedly split at random into A, B and an evaluation part (200 repetitions), and we count how often the selective risk, or the miss rate where a critical class exists, on the evaluation part exceeds ε. Paired accuracy differences are tested with the exact McNemar test (McNemar, 1947). Table complexity is the number of rules, the number of inputs and the mean number of non-trivial conditions per rule. Operational metrics are the model latency per case (median and 95th percentile), the time to evaluate the table, the cost per 1,000 cases at the published price of \$0.042 per million input tokens, and the number of cases per 1,000 sent to review.

4.6. Implementation

The method is implemented in Python. The DMN writer produces DMN 1.3 XML with diagram interchange information, so the models open in DMN modelling tools; the BPMN model uses Camunda 8 extensions. The code, the question schema, the identifiers of all sampled complaints with their splits, the generated DMN and BPMN models and the scripts that produce every number and figure of this paper are available in the repository (Section 7).

5. Results

5.1. RQ1: Fidelity and complexity of distilled tables

Table 4 summarises the distilled tables. For the relief decision the selected table has 2 rules over 1 inputs and reproduces Jev's holistic decision on 86.4% of the test cases (95% CI [84.8, 88.0]; $\kappa = 0.701$). For routing it has 31 rules over 11 inputs with a fidelity of 84.9% (95% CI [83.3, 86.4]; $\kappa = 0.825$). Merging reduced the tables from 2 and 156 tree leaves; with a FIRST hit policy and a default rule they need 2 and 29 rules. All tables passed every verification step: no conflicting overlaps and no gaps (as Lemma 1 predicts), XSD-valid XML, and identical outputs of the tree, our FEEL interpreter and the independent engine on all cases (round-trip agreement 100.0% and 100.0%). Figure 2 shows the fidelity–complexity trade-off over all configurations. Table 5 shows the relief table as it appears in the DMN model.

Table 4
Distilled business decision tables (test set). Fidelity is agreement with Jev's holistic decision; overlaps/gaps after merging.

	Rules							
Task	Tree	Merged	FIRST	Inputs	Cond./rule	Fidelity (%) [95% CI]	κ	Overl./gaps
Monetary relief	2	2	2	1	1.00	86.4 [84.8, 88.0]	0.701	0/0
Complaint routing	156	31	29	11	3.61	84.9 [83.3, 86.4]	0.825	0/0
Job-posting fraud	93	49	29	11	5.61	86.9 [85.7, 88.1]	0.451	0/0

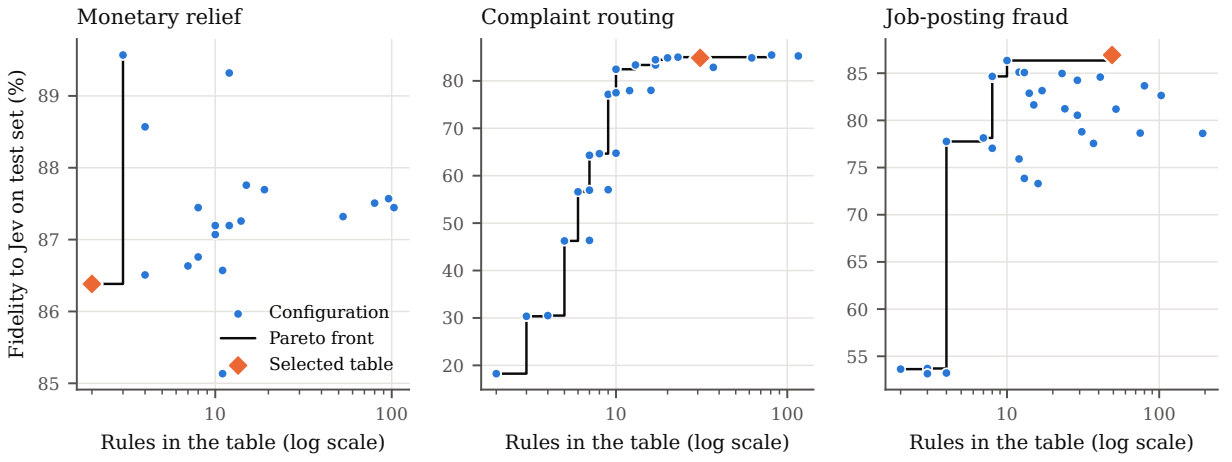


Figure 2: Fidelity to Jev against the number of rules for all configurations of the sweep, refitted on the pool and evaluated on the test set. The step line is the Pareto front; the diamond marks the selected table.

Table 5
Distilled decision table for monetary relief (inputs that are "-" in every rule omitted; support = pool cases per rule).

#	Money lost	Monetary relief	Support
1	`false`	`no_relief`	1808
2	`true`	`relief`	2991

Table 6
Calibration of Jev's holistic answer against the ground truth (test set).

Task	Acc. (%)	Mean conf. (%)	ECE	ECE_q	MCE	Brier	ECE (isotonic)
Monetary relief	48.3	74.5	0.261	0.261	0.341	0.628	0.045
Complaint routing	70.8	91.1	0.203	0.203	0.362	0.481	0.016
Job-posting fraud	86.8	74.1	0.129	0.127	0.158	0.234	0.011

5.2. RQ2: Calibration

Table 6 and Figure 3 report how well Jev's holistic probabilities match the true outcomes. On the test set, Jev's holistic answer is correct for 48.3% of relief cases at a mean confidence of 74.5% (ECE 0.261, Brier 0.628) and for 70.8% of routing cases at a mean confidence of 91.1% (ECE 0.203, Brier 0.481). Isotonic recalibration on the calibration set changes the ECE to 0.045 and 0.016, respectively.

5.3. RQ3: Automation under a risk constraint

Table 7 reports, for each gate and risk level, the share of test cases decided automatically and the error rate among them; Figure 4 shows the full risk–coverage curves. The table agrees with Jev's holistic answer

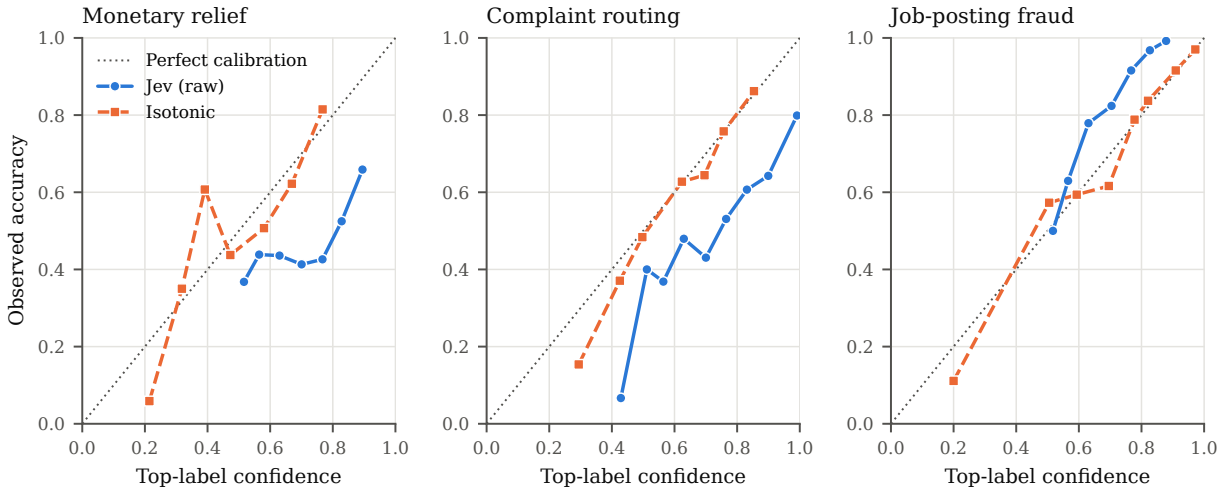


Figure 3: Reliability diagrams of Jev's top-label confidence on the test set, raw and after isotonic recalibration (bins with at least 10 cases).

Table 7
Coverage (Cov., % of test cases automated) and selective risk (Risk, % errors among automated cases) of the confidence gates, $\delta = 0.05$. "–" marks levels at which no threshold was certified and all cases go to review.

		$\varepsilon = 0.05$			$\varepsilon = 0.1$		
Task	Gate (with agreement filter)	Cov.	Risk	Miss	Cov.	Risk	Miss
Monetary relief	Jev confidence	0.0	–	–	0.0	–	–
	Learned gate	0.0	–	–	0.0	–	–
	Learned gate, DMN-Sup	0.0	–	–	0.0	–	–
Complaint routing	Jev confidence	0.0	–	n/a	0.0	–	n/a
	Learned gate	0.0	–	n/a	40.5	7.8	n/a
	Learned gate, DMN-Sup	0.0	–	n/a	51.6	9.2	n/a
Job-posting fraud	Jev confidence	0.0	–	–	0.0	–	–
	Learned gate	0.0	–	–	0.0	–	–
	Learned gate, DMN-Sup	0.0	–	–	12.7	0.8	0.8

on 86.4% of relief and 84.9% of routing test cases; when it agrees, it is correct on 47.1% and 76.9% of the cases, and when it disagrees on 44.0% and 46.9%. At $\varepsilon = 0.05$, the learned gate with the agreement filter automates 0.0% of relief cases with an error rate of 0.0%, and 0.0% of routing cases with an error rate of 0.0%. The AURC of the learned gate is 0.338 (relief) and 0.151 (routing), against 0.429 and 0.186 for Jev's confidence. In the repeated-splitting check, the selective risk exceeded $\varepsilon = 0.05$ in 0.0% (relief) and 0.0% (routing) of the 200 repetitions, against the nominal $\delta = 5\%$.

5.4. RQ4: Comparison with baselines and operational cost

Table 8 compares the approaches. Without any labels, the distilled table reaches an accuracy of 46.7% on relief and 72.4% on routing, against 48.3% and 70.8% for Jev's holistic answer (exact McNemar test: $p = 0.0902$ and $p = 0.0591$). With all 4,799 (relief) and 6,000 (routing) pool labels, the supervised table reaches 61.1% and 75.0%, and TF-IDF+LR 83.1% and 74.0%. Figure 5 shows how the supervised baselines depend on the number of labels.

The model answered a request with all sixteen questions in a median of 283 ms (95th percentile 430 ms) at a mean of 1454 input tokens, which corresponds to $0.0611 per 1,000 cases. Evaluating the distilled table took 0.4 μs per case in our Python implementation. At $\varepsilon = 0.05$ the relief process sends 1000 and the routing process 1000 of every 1,000 cases to review.

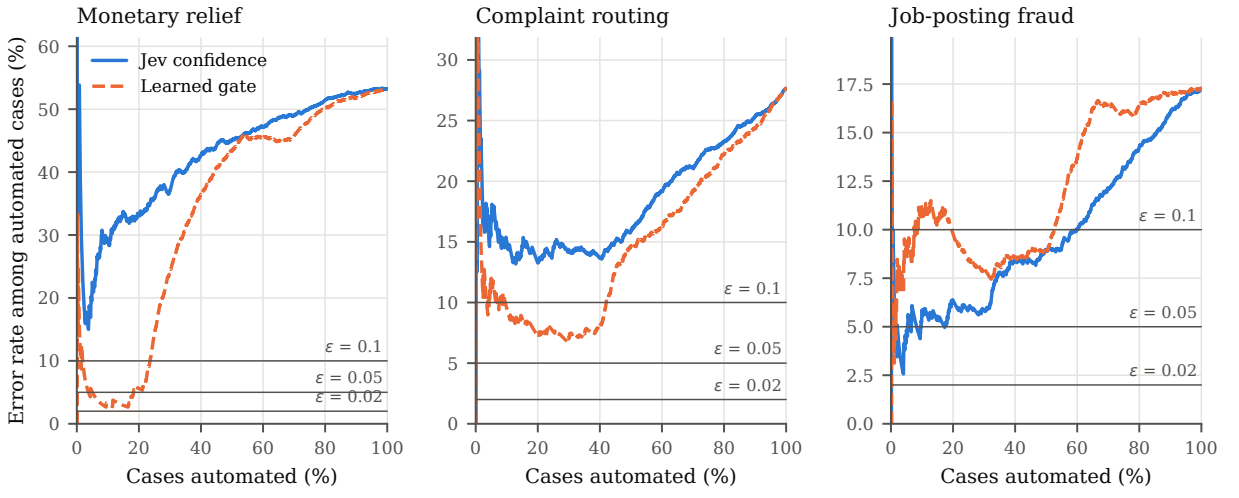


Figure 4: Risk–coverage curves on the test set (agreement filter not applied), with the risk levels ε used for calibration.

Table 8
Accuracy against ground truth on the test set. Labels = ground-truth labels used for training; supervised results with all pool labels.

Task	Method	Labels	Acc. (%)	Bal. acc. (%)	Recall$_+$ (%)	Auditable logic
Monetary relief	Majority class	4,799	83.2	50.0	0.0	–
	Jev holistic	0	48.3	64.7	89.2	no
	Jev + DMN, distilled (ours)	0	46.7	61.9	84.8	yes
	Jev + DMN, supervised (DMN-Sup)	4,799	61.1	66.7	75.1	yes
	TF-IDF + logistic regression	4,799	83.1	52.1	5.6	no
Complaint routing	Majority class	6,000	12.5	12.5	–	–
	Jev holistic	0	70.8	70.8	–	no
	Jev + DMN, distilled (ours)	0	72.4	72.4	–	yes
	Jev + DMN, supervised (DMN-Sup)	6,000	75.0	75.0	–	yes
	TF-IDF + logistic regression	6,000	74.0	74.0	–	no
Job-posting fraud	Majority class	8,736	95.8	50.0	0.0	–
	Jev holistic	0	86.8	61.2	33.3	no
	Jev + DMN, distilled (ours)	0	82.7	58.3	31.7	yes
	Jev + DMN, supervised (DMN-Sup)	8,736	78.6	76.0	73.2	yes
	TF-IDF + logistic regression	8,736	97.3	67.5	35.0	no

6. Discussion

6.1. Implications for research

The method separates three concerns that black-box automation entangles: reading the case, deciding, and deciding whether to decide. Reading is delegated to a model with typed output; deciding is done by a table that a person can read and that an engine executes; whether to decide is governed by a second table whose threshold carries a statistical guarantee. Each part can be inspected and replaced independently. This extends decision discovery (Bazhenova et al., 2016; De Smedt et al., 2019; Etinger et al., 2019) from structured logs to unstructured input, and it turns the abstract idea of a concept bottleneck (Koh et al., 2020) into an artefact of an industry standard.

Distillation from a model with typed output avoids two problems of distilling from free-text LLM output. There is nothing to parse: every answer is a value of a declared type, so the student's inputs are well defined.

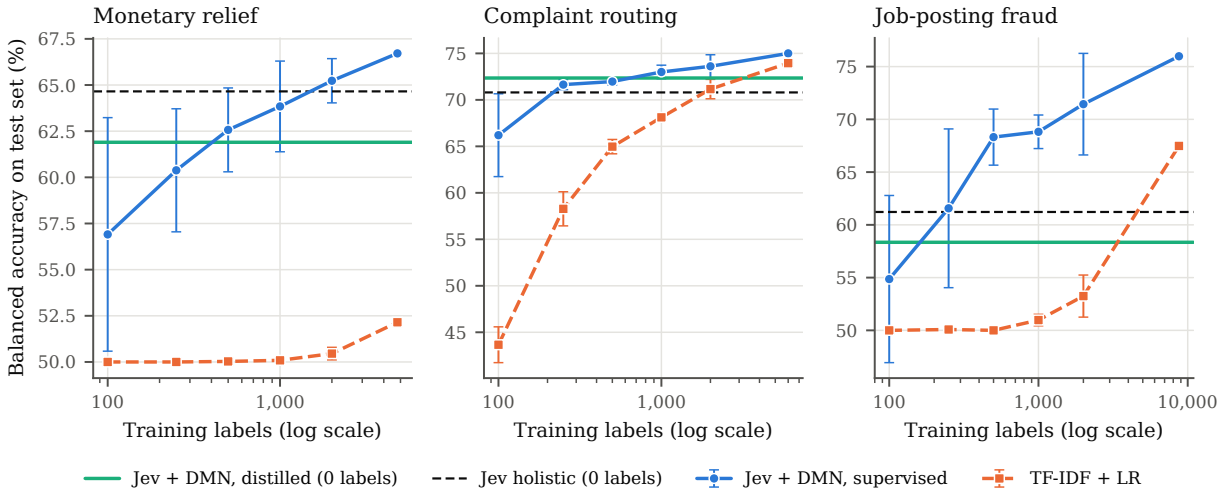


Figure 5: Test accuracy of supervised baselines against the number of training labels (mean ± standard deviation over five draws), compared with the two label-free approaches.

And the teacher's labels come with probabilities, which a text-generating model provides only through token likelihoods or verbalised confidence (Kadavath et al., 2022; Xiong et al., 2024). Fidelity measures how much of the model's holistic judgement can be expressed through the chosen facts; low fidelity is itself informative, because it signals that the schema misses a factor the model uses.

The confidence gate uses calibration data only where it is needed. Proposition 1 holds whether or not the model is calibrated, so a process owner can rely on the stated error bound even for a model whose calibration on the task is unknown. Calibration, which RQ2 measures, determines how many cases the bound allows to be automated.

6.2. Implications for practice

For organisations, the approach provides a path to automating decisions on unstructured input that fits existing governance. The decision logic lives in a DMN table that business analysts can review and change through the usual change process; the knowledge sources in the DRD document which model version and which calibration data the rules come from. A risk owner sets ε per decision, as with the different thresholds for low- and high-stakes actions in the vendor's guidance (TypeSafe AI, 2026c), and the gate shows the price of that choice in reviewer workload. Every automated decision can be explained by the rule that fired and the facts that matched, which supports the transparency and human-oversight obligations of the AI Act (European Parliament and Council of the European Union, 2024). Because the model is called once per case and the table evaluation is negligible, the latency and the cost per case are those of a single model request.

6.3. Limitations and threats to validity

Construct validity. The ground truth of the relief task is what companies did, not what they should have done; relief also depends on company policies and on facts not in the narrative. The routing ground truth is the consumer's own choice of product and issue, which is a proxy for the correct queue. Both are noisy and bound the achievable accuracy of every method.

Internal validity. Splits are stratified and fixed by a seed, the pool's labels are never used by the distilled table, calibration and test sets are disjoint, and exact duplicate narratives are removed. Near-duplicates may remain. The schema was designed before the experiments and was not tuned on the test set; a different schema would lead to different tables, and schema design is part of the method rather than a free parameter.

External validity. We study one domain, one language and one model. Jev is a commercial model in early access whose weights are not available, so its results cannot be reproduced bit for bit; we mitigate this by pinning the model version, storing every request and response, and releasing the complaint identifiers. The method only assumes typed questions with probability distributions over answers, so it applies to other

models of the class. An open adapter that emulates the System One interface with LLMs (TypeSafe AI, 2026b) makes a reproduction with open models possible, at different latency and cost. The narratives come from a public mirror because the CFPB withdrew them from its database; the identifiers and the mirror's checksums are released so that the sample can be reconstructed.

Statistical validity. Proposition 1 assumes that future cases are exchangeable with the calibration cases. Changes in the case mix, the company policy or the model version break this assumption and require recalibration, for which the process supplies the labels.

7. Conclusion

We showed how a System One model, which answers typed questions with probability distributions, can be turned into an auditable decision service in DMN. The model teaches a decision table over its own typed answers, which is verified and executed by a standard engine, and its probabilities drive a confidence gate whose threshold bounds the error rate of automated cases, and the miss rate of a critical class, with a chosen probability. On three decisions over CFPB complaints and EMSCAD job postings we measured the fidelity and size of the distilled tables, the calibration of the model, the share of cases that can be automated at a given risk, and the trade-offs against the model alone and against supervised alternatives. Future work will apply the method to processes with richer state, combine distillation with the labels that reviewers produce, and study how tables and gates should be maintained when the underlying model is updated.

CRediT authorship contribution statement

CRediT authorship contribution statement

Darko Etinger: Conceptualization, Methodology, Software, Investigation, Writing – original draft, Writing – review & editing. **Nikola Tanković:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Claude (Anthropic) to assist with software development and with drafting and editing the manuscript. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Data availability

Code, question schema, complaint identifiers with splits, generated DMN and BPMN models, raw model responses and analysis scripts will be made available at [repository URL]. The complaint data are public (Consumer Financial Protection Bureau, 2026; Heineman, 2025).

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References

Angelopoulos, A.N., Bates, S., Candès, E.J., Jordan, M.I., Lei, L., 2025. Learn then test: Calibrating predictive algorithms to achieve risk control. The Annals of Applied Statistics 19. doi:10.1214/24-aoas1998.

Angelopoulos, A.N., Bates, S., Fisch, A., Lei, L., Schuster, T., 2024. Conformal risk control, in: The Twelfth International Conference on Learning Representations (ICLR).

Arco, L., Nápoles, G., Vanhoenshoven, F., Lara, A.L., Casas, G., Vanhoof, K., 2021. Natural language techniques supporting decision modelers. Data Mining and Knowledge Discovery 35, 290–320. doi:10.1007/s10618-020-00718-4.

Ballard Spahr LLP, 2026. CFPB ends publication of consumer complaint narratives and data visualizations. Consumer Finance Monitor, 18 August 2026, https://www.consumerfinancemonitor.com/2026/08/18/cfpb-ends-publication-of-consumer-complaint-narratives-and-data-visualizations/.

Barredo Arrieta, A., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., Garcia, S., Gil-Lopez, S., Molina, D., Benjamins, R., Chatila, R., Herrera, F., 2020. Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. Information Fusion 58, 82–115. doi:10.1016/j.inffus.2019.12.012.

Bastani, K., Namavari, H., Shaffer, J., 2019. Latent Dirichlet allocation (LDA) for topic modeling of the CFPB consumer complaints. Expert Systems with Applications 127, 256–271. doi:10.1016/j.eswa.2019.03.001.

Bastani, O., Kim, C., Bastani, H., 2017. Interpreting blackbox models via model extraction. arXiv:1705.08504.

Batoulis, K., Meyer, A., Bazhenova, E., Decker, G., Weske, M., 2015. Extracting Decision Logic from Process Models, in: Advanced Information Systems Engineering, Springer International Publishing. pp. 349–366. doi:10.1007/978-3-319-19069-3_22.

Bazhenova, E., Buelow, S., Weske, M., 2016. Discovering Decision Models from Event Logs, in: Business Information Systems, Springer International Publishing. pp. 237–251. doi:10.1007/978-3-319-39426-8_19.

Benara, V., Singh, C., Morris, J.X., Antonello, R.J., Stoica, I., Huth, A.G., Gao, J., 2024. Crafting interpretable embeddings for language neuroscience by asking LLMs questions, in: Advances in Neural Information Processing Systems 37 (NeurIPS 2024). doi:10.52202/079017-3944.

Bork, D., Ali, S.J., Dinev, G.M., 2023. AI-Enhanced Hybrid Decision Management. Business & Information Systems Engineering 65, 179–199. doi:10.1007/s12599-023-00790-2.

BRIER, G.W., 1950. Verification of forecasts expressed in terms of probability. Monthly Weather Review 78, 1–3. doi:10.1175/1520-0493(1950)078<0001:vofeit>2.0.co;2.

Calvanese, D., Dumas, M., Laurson, Ü., Maggi, F.M., Montali, M., Teinemaa, I., 2016. Semantics and Analysis of DMN Decision Tables, in: Business Process Management, Springer International Publishing. pp. 217–233. doi:10.1007/978-3-319-45348-4_13.

Calvanese, D., Dumas, M., Laurson, Ü., Maggi, F.M., Montali, M., Teinemaa, I., 2018. Semantics, Analysis and Simplification of DMN Decision Tables. Information Systems 78, 112–125. doi:10.1016/j.is.2018.01.010.

Cherednichenko, O., Maliarenko, V., 2025. AI-Based Efficient Automation of Decision Logic Representation, in: Smart Business Technologies, Springer Nature Switzerland. pp. 303–315. doi:10.1007/978-3-032-08614-3_19.

Chow, C., 1970. On optimum recognition error and reject tradeoff. IEEE Transactions on Information Theory 16, 41–46. doi:10.1109/tit.1970.1054406.

Cohen, J., 1960. A Coefficient of Agreement for Nominal Scales. Educational and Psychological Measurement 20, 37–46. doi:10.1177/001316446002000104.

Consumer Financial Protection Bureau, 2026. Consumer Complaint Database. https://www.consumerfinance.gov/data-research/consumer-complaints/. Bulk snapshot of 6 October 2026.

Craven, M.W., Shavlik, J.W., 1996. Extracting tree-structured representations of trained networks, in: Advances in Neural Information Processing Systems 8, MIT Press.

De Smedt, J., Hasić, F., vanden Broucke, S.K., Vanthienen, J., 2019. Holistic discovery of decision models from process execution data. Knowledge-Based Systems 183, 104866. doi:10.1016/j.knosys.2019.104866.

Dumas, M., La Rosa, M., Mendling, J., Reijers, H.A., 2018. Fundamentals of Business Process Management. Springer Berlin Heidelberg. doi:10.1007/978-3-662-56509-4.

Etinger, D., Simić, S., Buljubašić, L., 2019. Automated decision-making with DMN: from decision trees to decision tables, in: 2019 42nd International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO), IEEE. pp. 1309–1313. doi:10.23919/mipro.2019.8756694.

European Parliament and Council of the European Union, 2024. Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union, L series, 12 July 2024. https://eur-lex.europa.eu/eli/reg/2024/1689/oj.

feelin contributors, 2026. feelin: A FEEL parser and interpreter. https://github.com/nikku/feelin. Version 8.2.0.

Figl, K., Mendling, J., Tokdemir, G., Vanthienen, J., 2018. What we know and what we do not know about DMN. Enterprise Modelling and Information Systems Architectures 13. doi:10.18417/EMISA.13.2.

Friedman, J.H., Popescu, B.E., 2008. Predictive learning via rule ensembles. The Annals of Applied Statistics 2. doi:10.1214/07-aoas148.

Frosst, N., Hinton, G., 2017. Distilling a neural network into a soft decision tree. arXiv:1711.09784.

Geifman, Y., El-Yaniv, R., 2017. Selective classification for deep neural networks, in: Advances in Neural Information Processing Systems 30.

Geng, J., Cai, F., Wang, Y., Koeppl, H., Nakov, P., Gurevych, I., 2024. A survey of confidence estimation and calibration in large language models, in: Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), Association for Computational Linguistics. pp. 6577–6595. doi:10.18653/v1/2024.naacl-long.366.

Gilardi, F., Alizadeh, M., Kubli, M., 2023. ChatGPT outperforms crowd workers for text-annotation tasks. Proceedings of the National Academy of Sciences 120. doi:10.1073/pnas.2305016120.

Goossens, A., De Smedt, J., Vanthienen, J., 2023. Extracting Decision Model and Notation models from text using deep learning techniques. Expert Systems with Applications 211, 118667. doi:10.1016/j.eswa.2022.118667.

Goossens, A., De Smedt, J., Vanthienen, J., 2024. Extracting process-aware decision models from object-centric process data. Information Sciences 682, 121263. doi:10.1016/j.ins.2024.121263.

Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F., Pedreschi, D., 2018. A Survey of Methods for Explaining Black Box Models. ACM Computing Surveys 51, 1–42. doi:10.1145/3236009.

Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q., 2017. On calibration of modern neural networks, in: Proceedings of the 34th International Conference on Machine Learning.

Hasić, F., De Smedt, J., Vanthienen, J., 2018. Augmenting processes with decision intelligence: Principles for integrated modelling. Decision Support Systems 107, 1–12. doi:10.1016/j.dss.2017.12.008.

Heineman, D., 2025. CFPB complaints (consumer-finance-complaints-large). Hugging Face dataset, https://huggingface.co/datasets/davidheineman/consumer-finance-complaints-large. Complaints from 1 December 2011 to 2 January 2025.

Hinton, G., Vinyals, O., Dean, J., 2015. Distilling the knowledge in a neural network. arXiv:1503.02531.

Horner, L.F., Moller, M., Reichert, M., 2026. Automated Business Process Model Generation With Large Language Models: A Systematic Literature Review. IEEE Access 14, 125550–125568. doi:10.1109/access.2026.3723936.

Kadavath, S., et al., 2022. Language models (mostly) know what they know. arXiv:2207.05221.

Kahneman, D., 2011. Thinking, Fast and Slow. Farrar, Straus and Giroux, New York.

Kapoor, S., Gruver, N., Roberts, M., Collins, K., Pal, A., Bhatt, U., Weller, A., Dooley, S., Goldblum, M., Wilson, A.G., 2024. Large language models must be taught to know what they don't know, in: Advances in Neural Information Processing Systems 37 (NeurIPS 2024).

Koh, P.W., Nguyen, T., Tang, Y.S., Mussmann, S., Pierson, E., Kim, B., Liang, P., 2020. Concept bottleneck models, in: Proceedings of the 37th International Conference on Machine Learning.

Leewis, S., Smit, K., Versendaal, J., 2024. Discovering operational decisions from data—a framework supporting decision discovery from data. DECISION 51, 417–436. doi:10.1007/s40622-024-00402-2.

de Leoni, M., van der Aalst, W.M.P., 2013. Data-aware process mining: discovering decisions in processes using alignments, in: Proceedings of the 28th Annual ACM Symposium on Applied Computing, ACM. pp. 1454–1461. doi:10.1145/2480362.2480633.

Madras, D., Pitassi, T., Zemel, R., 2018. Predict responsibly: Improving fairness and accuracy by learning to defer, in: Advances in Neural Information Processing Systems 31.

McNemar, Q., 1947. Note on the Sampling Error of the Difference Between Correlated Proportions or Percentages. Psychometrika 12, 153–157. doi:10.1007/bf02295996.

Mozannar, H., Sontag, D., 2020. Consistent estimators for learning to defer to an expert, in: Proceedings of the 37th International Conference on Machine Learning.

Object Management Group, 2021. Decision Model and Notation (DMN), Version 1.3. https://www.omg.org/spec/DMN/1.3/.

Pakdaman Naeini, M., Cooper, G., Hauskrecht, M., 2015. Obtaining Well Calibrated Probabilities Using Bayesian Binning. Proceedings of the AAAI Conference on Artificial Intelligence 29. doi:10.1609/aaai.v29i1.9602.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, É., 2011. Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12, 2825–2830.

Quach, V., Fisch, A., Schuster, T., Yala, A., Sohn, J.H., Jaakkola, T., Barzilay, R., 2024. Conformal language modeling, in: The Twelfth International Conference on Learning Representations (ICLR 2024).

Raghu, M., Blumer, K., Corrado, G., Kleinberg, J., Obermeyer, Z., Mullainathan, S., 2019. The algorithmic automation problem: Prediction, triage, and human effort. arXiv:1903.12220.

Ren, A.Z., Dixit, A., Bodrova, A., Singh, S., Tu, S., Brown, N., Xu, P., Takayama, L., Xia, F., Varley, J., Xu, Z., Sadigh, D., Zeng, A., Majumdar, A., 2023. Robots that ask for help: uncertainty alignment for large language model planners, in: Proceedings of The 7th Conference on Robot Learning, pp. 661–682.

Ribeiro, M.T., Singh, S., Guestrin, C., 2016. "Why Should I Trust You?": Explaining the Predictions of Any Classifier, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, ACM. pp. 1135–1144. doi:10.1145/2939672.2939778.

Rozinat, A., van der Aalst, W.M.P., 2006. Decision Mining in ProM, in: Business Process Management, Springer Berlin Heidelberg. pp. 420–425. doi:10.1007/11841760_33.

Rudin, C., 2019. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nature Machine Intelligence 1, 206–215. doi:10.1038/s42256-019-0048-x.

Simić, S.D., Tanković, N., Etinger, D., 2020. Automated decision modeling with DMN and BPMN: a model ensemble approach, in: Ahram, T., Karwowski, W., Pickl, S., et al. (Eds.), Human Systems Engineering and Design II, Springer, Cham. pp. 789–794. doi:10.1007/978-3-030-27928-8_120.

Singh, C., Askari, A., Caruana, R., Gao, J., 2023. Augmenting interpretable models with large language models during training. Nature Communications 14, 7913. doi:10.1038/s41467-023-43713-1.

Sun, C.E., Oikarinen, T., Ustun, B., Weng, T.W., 2025. Concept bottleneck large language models, in: The Thirteenth International Conference on Learning Representations (ICLR 2025).

Tian, K., Mitchell, E., Zhou, A., Sharma, A., Rafailov, R., Yao, H., Finn, C., Manning, C., 2023. Just Ask for Calibration: Strategies for Eliciting Calibrated Confidence Scores from Language Models Fine-Tuned with Human Feedback, in: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics. pp. 5433–5442. doi:10.18653/v1/2023.emnlp-main.330.

TypeSafe AI, 2026a. Jev 1.13 jaggedness. https://docs.typesafe.ai/model-jaggedness/jev-1.13. Last reviewed 2 October 2026; accessed 7 October 2026.

TypeSafe AI, 2026b. system-one-adapter-python: a System One interface for LLM providers. https://github.com/typesafe-ai/system-one-adapter-python. Accessed 7 October 2026.

TypeSafe AI, 2026c. TypeSafe documentation: System One API, primitives, confidence and models. https://docs.typesafe.ai. Accessed 7 October 2026.

Vidgof, M., Bachhofner, S., Mendling, J., 2023. Large Language Models for Business Process Management: Opportunities and Challenges, in: Business Process Management Forum, Springer Nature Switzerland. pp. 107–123. doi:10.1007/978-3-031-41623-1_7.

Vidros, S., Kolias, C., Kambourakis, G., Akoglu, L., 2017. Automatic detection of online recruitment frauds: characteristics, methods, and a public dataset. Future Internet 9, 6. doi:10.3390/fi9010006.

Vovk, V., Gammerman, A., Shafer, G., 2022. Algorithmic Learning in a Random World. Springer International Publishing. doi:10.1007/978-3-031-06649-8.

Wang, S., Liu, Y., Xu, Y., Zhu, C., Zeng, M., 2021. Want To Reduce Labeling Cost? GPT-3 Can Help, in: Findings of the Association for Computational Linguistics: EMNLP 2021, Association for Computational Linguistics. pp. 4195–4205. doi:10.18653/v1/2021.findings-emnlp.354.

Wen, B., Yao, J., Feng, S., Xu, C., Tsvetkov, Y., Howe, B., Wang, L.L., 2025. Know your limits: a survey of abstention in large language models. Transactions of the Association for Computational Linguistics 13, 529–556. doi:10.1162/tacl_a_00754.

Xiong, M., Hu, Z., Lu, X., Li, Y., Fu, J., He, J., Hooi, B., 2024. Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs, in: The Twelfth International Conference on Learning Representations (ICLR).

Zadrozny, B., Elkan, C., 2002. Transforming classifier scores into accurate multiclass probability estimates, in: Proceedings of the eighth ACM SIGKDD international conference on Knowledge discovery and data mining, ACM. pp. 694–699. doi:10.1145/775047.775151.